Academic_pressure

February 5, 2026

1 CALLING LIBRARY

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2 LOAD DATA

[2]:
```
df1 = pd.read_csv("d1.csv")
```

3 INFO FOR DATA

[3]:
```
df1.sample(3)
```

[3]:
```
              Timestamp Your Academic Stage  Peer pressure  \
89  25/07/2025 20:37:52       post-graduate              2
44  24/07/2025 22:34:34       undergraduate              4
2   24/07/2025 22:06:39       undergraduate              1

    Academic pressure from your home Study Environment  \
89                                 2             Noisy
44                                 2             Noisy
2                                  1          Peaceful

           What coping strategy you use as a student?  \
89  Analyze the situation and handle it with intel…
44  Analyze the situation and handle it with intel…
2                      Social support (friends, family)

   Do you have any bad habits like smoking, drinking on a daily basis?  \
89                                                 No
44                                                 No
2                                                  No

    What would you rate the academic  competition in your student life  \
```

```
89                                                    4
44                                                    2
2                                                     2

    Rate your academic stress index
89                                  4
44                                  2
2                                   4
```

[4]:
```
# DISPLAY NAN VALUES
```

4 cleaning data

[5]:
```
df1.isna().sum()
```

[5]:
```
Timestamp                                                            0
Your Academic Stage                                                  0
Peer pressure                                                        0
Academic pressure from your home                                     0
Study Environment                                                    1
What coping strategy you use as a student?                           0
Do you have any bad habits like smoking, drinking on a daily basis?  0
What would you rate the academic  competition in your student life   0
Rate your academic stress index                                      0
dtype: int64
```

[6]:
```
# DROP ROW INCLUDE NAN
```

[7]:
```
df1=df1.dropna()
```

[8]:
```
# DISPLAY COLUMN NAMES
```

[9]:
```
df1.columns
```

[9]:
```
Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
       'Academic pressure from your home', 'Study Environment',
       'What coping strategy you use as a student?',
       'Do you have any bad habits like smoking, drinking on a daily basis?',
       'What would you rate the academic  competition in your student life',
       'Rate your academic stress index '],
      dtype='object')
```

[10]:
```
# DISPLAY DATA TYPE FOR EVERY COLUMN
```

[11]:
```
df1.dtypes
```

[11]:
```
Timestamp                                                               object
Your Academic Stage                                                     object
Peer pressure                                                            int64
Academic pressure from your home                                         int64
Study Environment                                                       object
What coping strategy you use as a student?                              object
Do you have any bad habits like smoking, drinking on a daily basis?     object
What would you rate the academic  competition in your student life       int64
Rate your academic stress index                                          int64
dtype: object
```

[12]:
```
# DISPLAY SIMPLE Description FOR DATA
```

[13]:
```
df1.describe().round()
```

[13]:
```
       Peer pressure  Academic pressure from your home  \
count          139.0                             139.0
mean             3.0                               3.0
std              1.0                               1.0
min              1.0                               1.0
25%              2.0                               2.0
50%              3.0                               3.0
75%              4.0                               4.0
max              5.0                               5.0

       What would you rate the academic  competition in your student life  \
count                                              139.0
mean                                                 3.0
std                                                  1.0
min                                                  1.0
25%                                                  3.0
50%                                                  4.0
75%                                                  4.0
max                                                  5.0

       Rate your academic stress index
count                            139.0
mean                               4.0
std                                1.0
min                                1.0
25%                                3.0
50%                                4.0
75%                                4.0
max                                5.0
```

[14]:
```
# DISPLAY GENERAL INFORMATION
```

[15]:
```
df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 139 entries, 0 to 139
Data columns (total 9 columns):
 #   Column                                                                    Non-
Null Count  Dtype
---  ------
--------------  -----
 0   Timestamp                                                                 139
non-null    object
 1   Your Academic Stage                                                       139
non-null    object
 2   Peer pressure                                                             139
non-null    int64
 3   Academic pressure from your home                                          139
non-null    int64
 4   Study Environment                                                         139
non-null    object
 5   What coping strategy you use as a student?                                139
non-null    object
 6   Do you have any bad habits like smoking, drinking on a daily basis?       139
non-null    object
 7   What would you rate the academic  competition in your student life        139
non-null    int64
 8   Rate your academic stress index                                           139
non-null    int64
dtypes: int64(4), object(5)
memory usage: 10.9+ KB
```

[ ]:
```

```

[ ]:
```

```

[16]:
```
# Identifying duplicate items
```

[17]:
```
df1["Your Academic Stage"].unique()
```

[17]:
```
array(['undergraduate', 'high school', 'post-graduate'], dtype=object)
```

[18]:
```
df1["Peer pressure"].unique()
```

[18]:
```
array([4, 3, 1, 5, 2])
```

[19]:
```
# Converting numbers into words that express meaning
```

[20]:
```
peer_pressure_map = {
    1: 'Very Low',
```

```
    2: 'Low',
    3: 'Moderate',
    4: 'High',
    5: 'Very High'
}

df1['Peer pressure'] = df1['Peer pressure'].map(peer_pressure_map)
```

[21]:
```
df1["Peer pressure"].unique()
```

[21]:
```
array(['High', 'Moderate', 'Very Low', 'Very High', 'Low'], dtype=object)
```

[22]:
```
df1["Academic pressure from your home"].unique()
```

[22]:
```
array([5, 4, 1, 2, 3])
```

[23]:
```
academic_map = {
    1: 'No Pressure',
    2: 'Slight Pressure',
    3: 'Moderate Pressure',
    4: 'Strong Pressure',
    5: 'Extreme Pressure'
}

#
df1['Academic pressure from your home'] = df1['Academic pressure from your␣
 ↪home'].map(academic_map)
```

[24]:
```
df1["Academic pressure from your home"].unique()
```

[24]:
```
array(['Extreme Pressure', 'Strong Pressure', 'No Pressure',
       'Slight Pressure', 'Moderate Pressure'], dtype=object)
```

[25]:
```
df1["Study Environment"].unique()
```

[25]:
```
array(['Noisy', 'Peaceful', 'disrupted'], dtype=object)
```

[26]:
```
df1["What coping strategy you use as a student?"].unique()
```

[26]:
```
array(['Analyze the situation and handle it with intellect',
       'Social support (friends, family)',
       'Emotional breakdown (crying a lot)'], dtype=object)
```

[27]:
```
coping_map = {
    'Analyze the situation and handle it with intellect': 'Problem Solving',
    'Social support (friends, family)': 'Social Support',
    'Emotional breakdown (crying a lot)': 'Emotional Release'
}
```

```python
df1['What coping strategy you use as a student?'] = df1['What coping strategy␣
↪you use as a student?'].map(coping_map)
```

[28]:
```python
# The column name was changed due to its length.
```

[29]:
```python
df1.rename(columns={'What coping strategy you use as a student?':␣
↪'Coping_Mechanism'}, inplace=True)
```

[30]:
```python
df1.sample()
```

[30]:
```
                Timestamp Your Academic Stage Peer pressure  \
131  12/08/2025 15:01:20         high school          High

    Academic pressure from your home Study Environment Coping_Mechanism  \
131                  Strong Pressure          Peaceful  Problem Solving

    Do you have any bad habits like smoking, drinking on a daily basis?  \
131                                                 No

     What would you rate the academic  competition in your student life  \
131                                                  3

     Rate your academic stress index
131                                4
```

[31]:
```python
df1.rename(columns={'Do you have any bad habits like smoking, drinking on a␣
↪daily basis?': 'Lifestyle_Habits'}, inplace=True)
```

[32]:
```python
df1["Lifestyle_Habits"].unique()
```

[32]:
```
array(['No', 'prefer not to say', 'Yes'], dtype=object)
```

[33]:
```python
# Substitution for clarification
```

[34]:
```python
df1["Lifestyle_Habits"]=df1["Lifestyle_Habits"].replace(["prefer not to␣
↪say"],"Decline",regex= True)
```

[35]:
```python
df1["Lifestyle_Habits"].unique()
```

[35]:
```
array(['No', 'Decline', 'Yes'], dtype=object)
```

[36]:
```python
df1.rename(columns={"What would you rate the academic  competition in your␣
↪student life": 'Academic Competition'}, inplace=True)
```

[37]:
```python
competition_map = {
    1: 'Negligible',
    2: 'Low',
```

```
    3: 'Moderate',
    4: 'High',
    5: 'Fierce'
}

df1['Academic Competition'] = df1['Academic Competition'].map(competition_map)
```

[38]:
```
df1['Academic Competition'].unique()
```

[38]:
```
array(['Moderate', 'Low', 'High', 'Fierce', 'Negligible'], dtype=object)
```

[39]:
```
df1.rename(columns={'Rate your academic stress index ':␣
↪'Academic_Stress_Level'}, inplace=True)

stress_map = {
    1: 'Minimal',
    2: 'Mild',
    3: 'Moderate',
    4: 'Severe',
    5: 'Overwhelming'
}

df1['Academic_Stress_Level'] = df1['Academic_Stress_Level'].map(stress_map)
```

[40]:
```
df1['Academic_Stress_Level'].unique()
```

[40]:
```
array(['Overwhelming', 'Moderate', 'Severe', 'Mild', 'Minimal'],
      dtype=object)
```

[ ]:
```

```

[ ]:
```

```

[41]:
```
# Breaking down the column of history into its primary components
```

[42]:
```
df1['Timestamp'] = pd.to_datetime(df1['Timestamp'], format='%d/%m/%Y %H:%M:%S',␣
↪utc=True)
df1['month'] = df1['Timestamp'].dt.month
df1['year'] = df1['Timestamp'].dt.year
df1['day_name'] = df1['Timestamp'].dt.day_name()
```

[43]:
```
df1['day'] = df1['Timestamp'].dt.day
```

[44]:
```
df1["day"]=df1["day"].astype(object)
```

[45]:
```
import calendar
```

[46]:
```
month_map = {i: calendar.month_name[i] for i in range(1, 13)}
df1['Month_Name'] = df1['month'].map(month_map)
```

[47]:
```
df1=df1.drop(["month"],axis=1)
```

[48]:
```
df1.sample()
```

[48]:
```
                    Timestamp Your Academic Stage Peer pressure  \
28 2025-07-24 22:19:51+00:00       undergraduate     Very High

   Academic pressure from your home Study Environment Coping_Mechanism  \
28                      No Pressure         disrupted   Social Support

   Lifestyle_Habits Academic Competition Academic_Stress_Level  year  \
28               No                 High          Overwhelming  2025

    day_name day Month_Name
28  Thursday  24       July
```

[49]:
```
df1["year"]=df1["year"].astype("object")
```

[50]:
```
df1.rename(columns = {"Academic pressure from your home":"Academic pressure␣
↪home"},inplace =True)
```

[51]:
```
df1
```

[51]:
```
                    Timestamp Your Academic Stage Peer pressure  \
0   2025-07-24 22:05:39+00:00       undergraduate          High
1   2025-07-24 22:05:52+00:00       undergraduate      Moderate
2   2025-07-24 22:06:39+00:00       undergraduate      Very Low
3   2025-07-24 22:06:45+00:00       undergraduate      Moderate
4   2025-07-24 22:08:06+00:00       undergraduate      Moderate
..                        …                 …           …
135 2025-08-17 13:02:04+00:00       undergraduate      Moderate
136 2025-08-18 14:36:00+00:00       undergraduate          High
137 2025-08-18 17:13:52+00:00       undergraduate      Moderate
138 2025-08-18 19:08:52+00:00       undergraduate          High
139 2025-08-18 22:40:13+00:00       undergraduate          High

    Academic pressure home Study Environment Coping_Mechanism  \
0         Extreme Pressure             Noisy  Problem Solving
1          Strong Pressure          Peaceful  Problem Solving
2              No Pressure          Peaceful   Social Support
3          Slight Pressure          Peaceful  Problem Solving
4        Moderate Pressure          Peaceful  Problem Solving
..                      …               …              …
135        Slight Pressure          Peaceful  Problem Solving
```

```
136        Slight Pressure         disrupted  Problem Solving
137      Moderate Pressure          Peaceful  Problem Solving
138       Extreme Pressure         disrupted   Social Support
139        Slight Pressure          Peaceful   Social Support

    Lifestyle_Habits Academic Competition Academic_Stress_Level  year  \
0                 No             Moderate           Overwhelming  2025
1                 No             Moderate               Moderate  2025
2                 No                  Low                 Severe  2025
3                 No                 High               Moderate  2025
4                 No                 High           Overwhelming  2025
..               …                    …                    …    …
135               No             Moderate                 Severe  2025
136               No             Moderate               Moderate  2025
137               No                  Low                 Severe  2025
138               No               Fierce           Overwhelming  2025
139               No                 High                 Severe  2025

     day_name day Month_Name
0    Thursday  24       July
1    Thursday  24       July
2    Thursday  24       July
3    Thursday  24       July
4    Thursday  24       July
..        …  ..        …
135    Sunday  17     August
136    Monday  18     August
137    Monday  18     August
138    Monday  18     August
139    Monday  18     August

[139 rows x 13 columns]
```

5 export new data

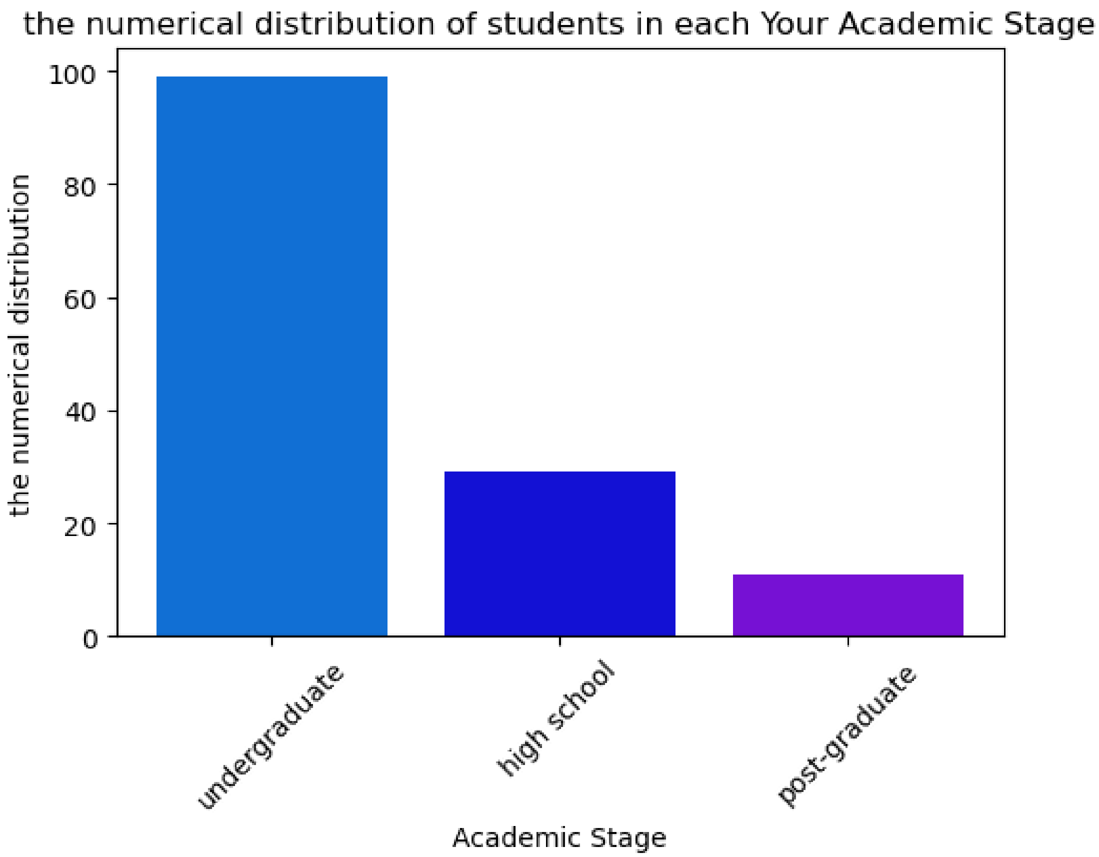
[52]:
```
# df1.to_csv("D:/projects/New folder/data/Edit_data/Academic_pressure.csv ")
```

6 analysis

6.0.1 What is the numerical distribution of students in each "Your Academic Stage"?

[53]:
```
g = df1.groupby("Your Academic Stage")["Your Academic Stage"].count().
 ↪sort_values(ascending=False)


plt.figure(figsize=(6,4))
```

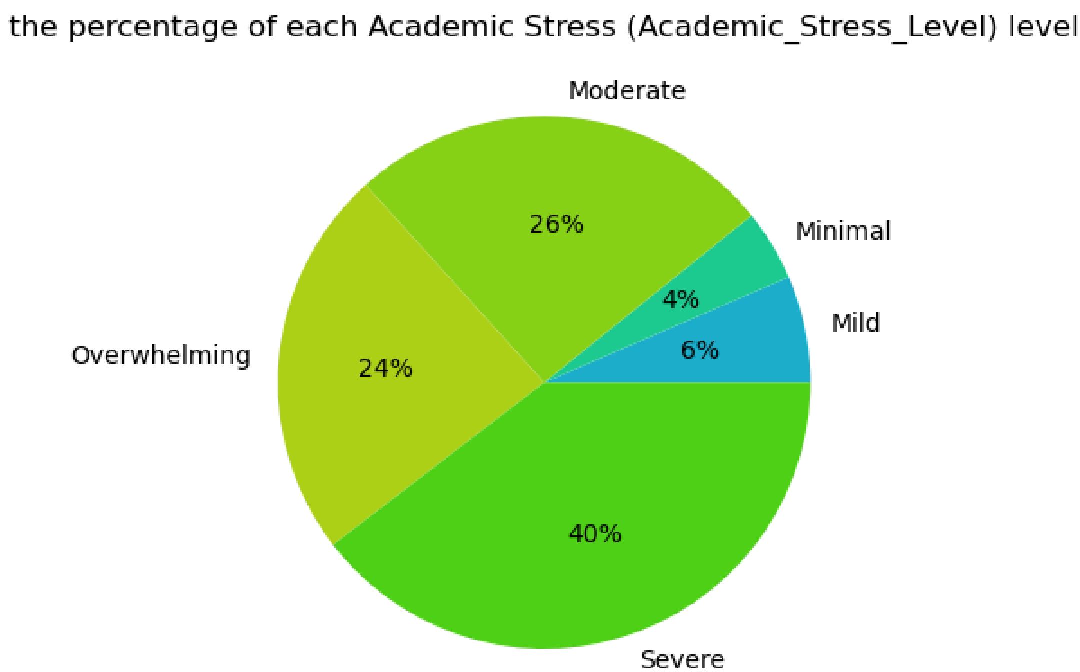
```
plt.bar(g.index,g.values,color=['#116FD4', '#1311D4', '#7611D4'])
plt.title("the numerical distribution of students in each Your Academic Stage")
plt.xlabel("Academic Stage")
plt.ylabel("the numerical distribution")
plt.xticks(rotation=45)
plt.show()
```



6.0.2 What is the percentage of each "Academic Stress" (Academic_Stress_Level) level in the sample?

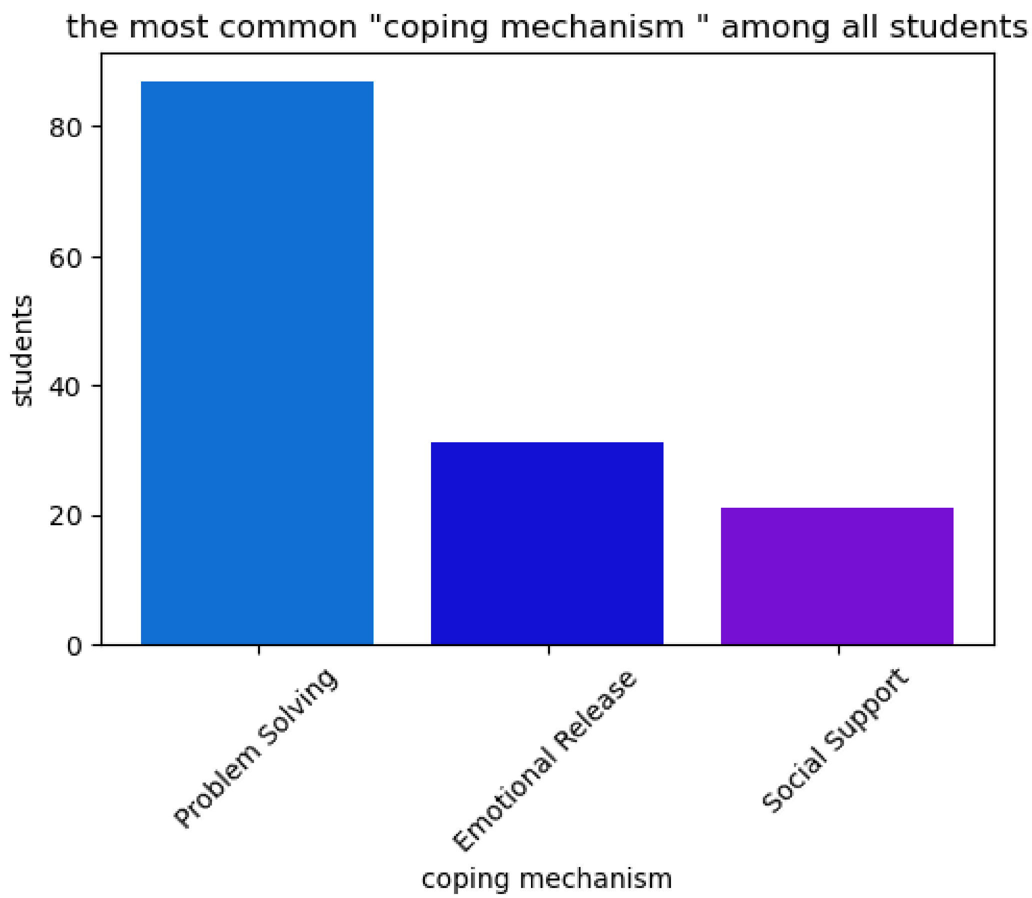
[54]:
```
j = df1.groupby("Academic_Stress_Level")["Academic_Stress_Level"].count()
plt.pie(
    j.values,
    labels= j.index,
    labeldistance=1.1,
    radius=1,
    colors=["#1CADCA","#1CCA90","#86D016","#ABD016","#4ED016"],
    rotatelabels= False,
    autopct= "%1.0f%%"
```

```
)
plt.title(" the percentage of each Academic Stress (Academic_Stress_Level)␣
↪level ")
plt.show()
```



6.0.3 What is the most common "coping mechanism" among all students?

[55]:
```
k = df1.groupby("Coping_Mechanism")["Coping_Mechanism"].count().
↪sort_values(ascending=False)


plt.figure(figsize=(6,4))
plt.bar(k.index,k.values,color=['#116FD4', '#1311D4', '#7611D4'])
plt.title("the most common \"coping mechanism \" among all students")
plt.xlabel("coping mechanism")
plt.ylabel("students")
plt.xticks(rotation=45)
plt.show()
```



6.0.4 Does Academic Stress Level increase with increased Academic pressure home?

[56]:
```
df1.columns
```

[56]:
```
Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
       'Academic pressure home', 'Study Environment', 'Coping_Mechanism',
       'Lifestyle_Habits', 'Academic Competition', 'Academic_Stress_Level',
       'year', 'day_name', 'day', 'Month_Name'],
      dtype='object')
```

[57]:
```
from scipy.stats import chi2_contingency

# Let's assume we want to study the relationship between the study environment␣
↪and Academic pressure home
contingency_table_one = pd.crosstab(df1['Study Environment'], df1['Academic␣
↪pressure home'],normalize='index')*100
print(contingency_table_one)
```

```
Academic pressure home  Extreme Pressure  Moderate Pressure  No Pressure  \
```

```
Study Environment
Noisy                        25.000000          25.000000    21.875000
Peaceful                     14.492754          36.231884     8.695652
disrupted                    23.684211          28.947368    10.526316

Academic pressure home  Slight Pressure  Strong Pressure
Study Environment
Noisy                          9.375000        18.750000
Peaceful                      24.637681        15.942029
disrupted                     10.526316        26.315789
```

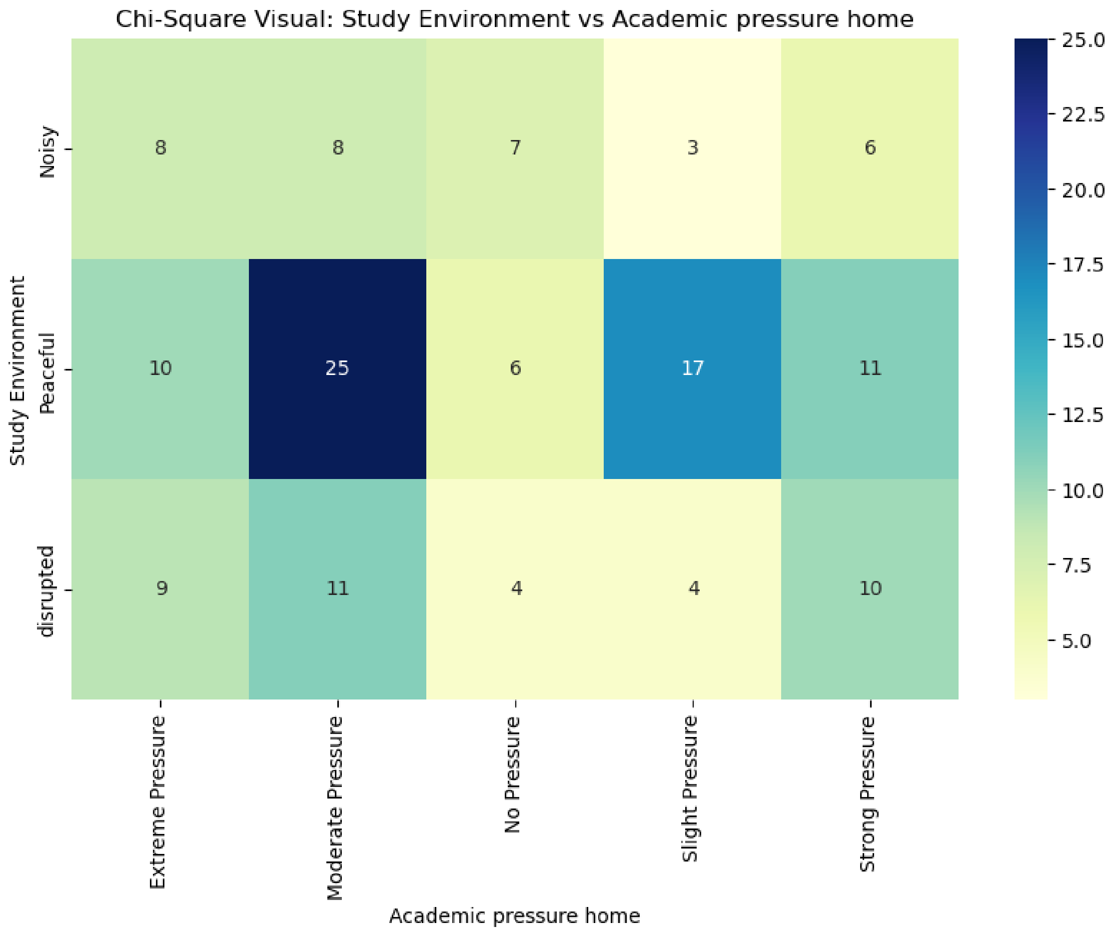
[72]:

```
#  Chi-Square
chi1, p1, dof1, expected1 = chi2_contingency(contingency_table_one)

print(f"Chi-Square Statistic: {chi1}")
print(f"P-value: {p1}")
print(f"{expected1}")
```

```
Chi-Square Statistic: 25.276576390718454
P-value: 0.0013953241259972685
[[21.05898805 30.05975083 13.69898932 14.84633232 20.33593949]
 [21.05898805 30.05975083 13.69898932 14.84633232 20.33593949]
 [21.05898805 30.05975083 13.69898932 14.84633232 20.33593949]]
```

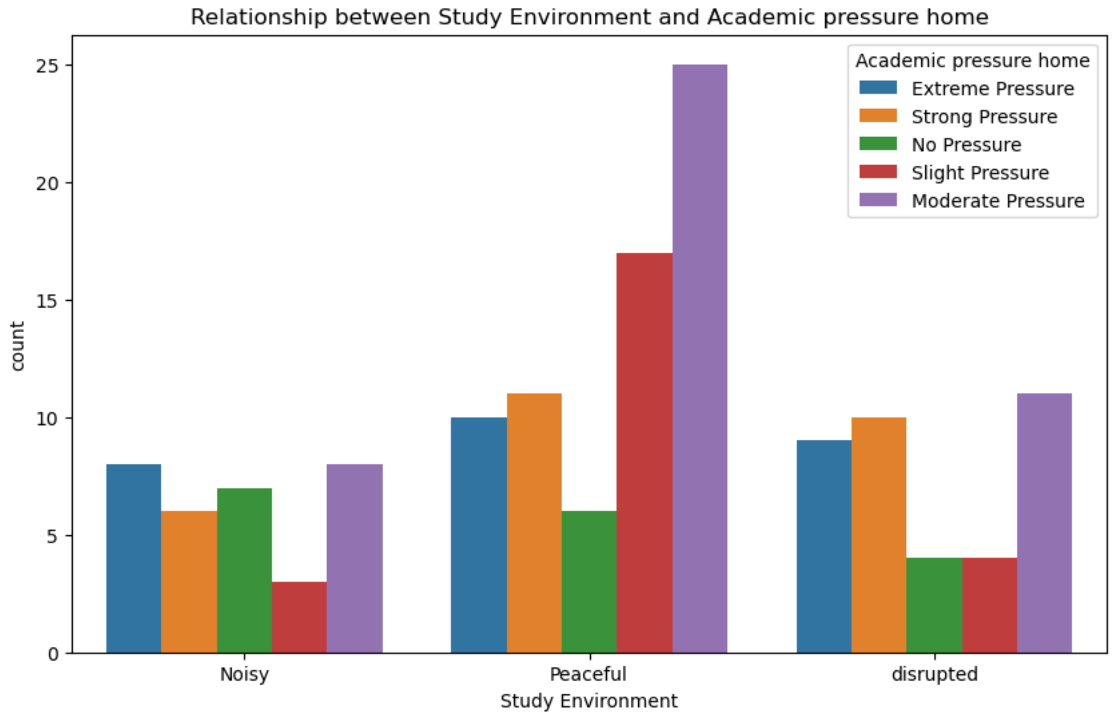
[59]:

```
# 1. Contingency Table
ct1 = pd.crosstab(df1['Study Environment'], df1['Academic pressure home'])

# 2. (Heatmap)
plt.figure(figsize=(10, 6))
sns.heatmap(ct1, annot=True, cmap='YlGnBu', fmt='d')

plt.title('Chi-Square Visual: Study Environment vs Academic pressure home')
plt.xlabel('Academic pressure home')
plt.ylabel('Study Environment')
plt.show()
```



[ ]:

[120]:
```
plt.figure(figsize=(10,6))
sns.countplot(data=df1, x='Study Environment', hue='Academic pressure home')
plt.title('Relationship between Study Environment and Academic pressure home')
plt.show()
```



[ ]:

6.0.5 What is the relationship between the "study environment" and stress levels? Does a calm environment actually mean less stress?

[61]:
```python
df1.columns
```

[61]:
```
Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
       'Academic pressure home', 'Study Environment', 'Coping_Mechanism',
       'Lifestyle_Habits', 'Academic Competition', 'Academic_Stress_Level',
       'year', 'day_name', 'day', 'Month_Name'],
      dtype='object')
```

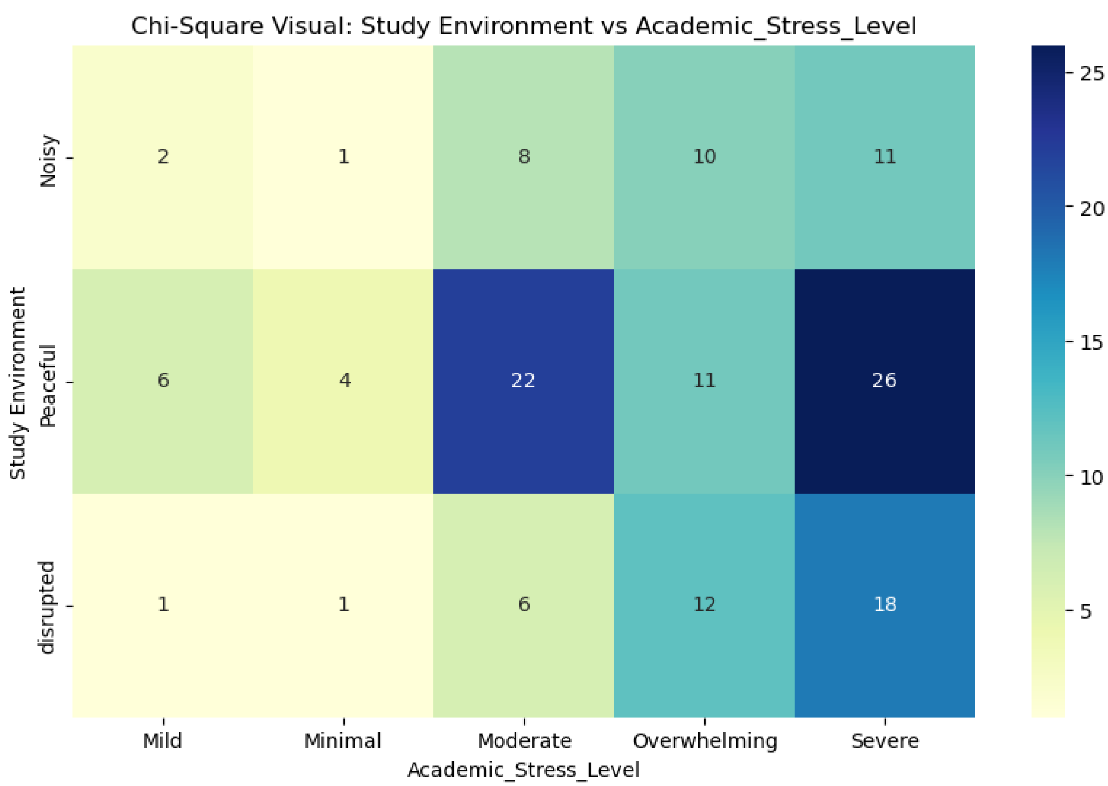
[62]:
```python
# Let's assume we want to study the relationship between the study environment␣
↪and stress levels.
contingency_table_two = pd.crosstab(df1['Study Environment'],␣
↪df1['Academic_Stress_Level'],normalize='index')*100
print(contingency_table_two)
```

```
Academic_Stress_Level      Mild   Minimal   Moderate  Overwhelming     Severe
Study Environment
Noisy                  6.250000  3.125000  25.000000     31.250000  34.375000
Peaceful               8.695652  5.797101  31.884058     15.942029  37.681159
disrupted              2.631579  2.631579  15.789474     31.578947  47.368421
```

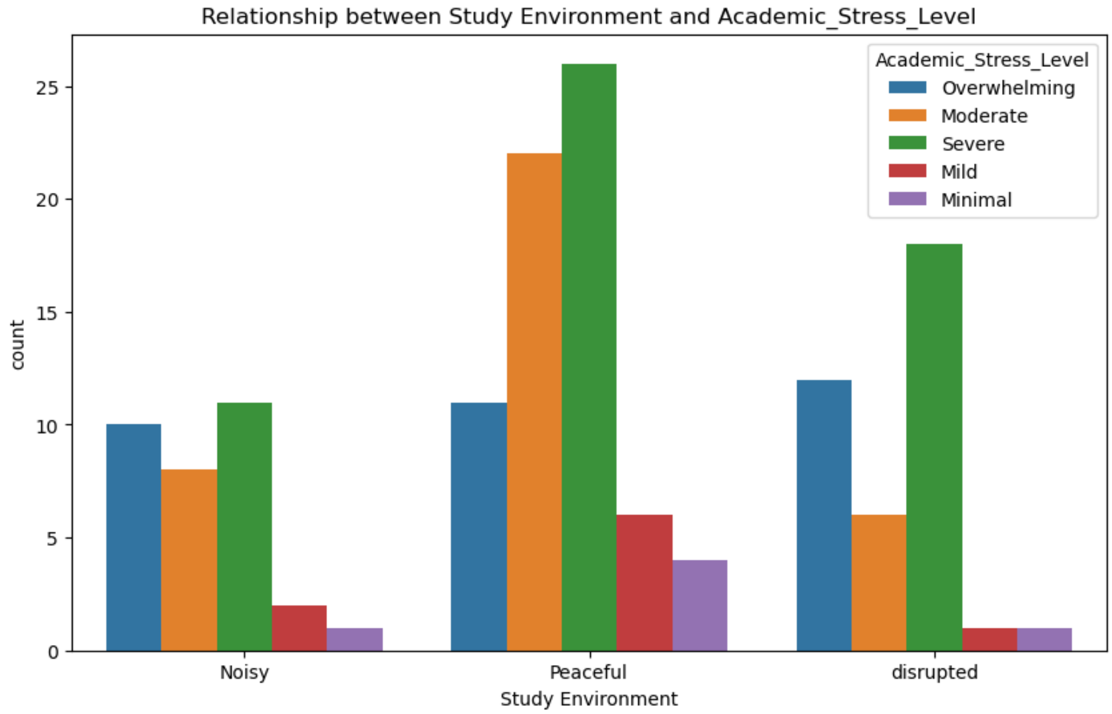
[63]:
```python
# Chi-Square
chi2, p2, dof2, expected2 = chi2_contingency(contingency_table_two)

print(f"Chi-Square Statistic: {chi2}")
print(f"P-value: {p2}")
```

```
Chi-Square Statistic: 18.438776696693054
P-value: 0.01816703201913433
```

[64]:
```python
# 1. (Contingency Table)
ct2 = pd.crosstab(df1['Study Environment'], df1['Academic_Stress_Level'])

# 2.              (Heatmap)
plt.figure(figsize=(10, 6))
sns.heatmap(ct2, annot=True, cmap='YlGnBu', fmt='d')

plt.title('Chi-Square Visual: Study Environment vs Academic_Stress_Level')
plt.xlabel('Academic_Stress_Level')
plt.ylabel('Study Environment')
plt.show()
```



[65]:
```
plt.figure(figsize=(10,6))
sns.countplot(data=df1, x='Study Environment', hue='Academic_Stress_Level')
plt.title('Relationship between Study Environment and Academic_Stress_Level')
plt.show()
```



[ ]:
```
```

[ ]:
```
```

6.0.6 Is there a relationship between "academic competition" and the emergence of certain "lifestyle habits"?

[66]:
```
df1.columns
```

[66]:
```
Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
       'Academic pressure home', 'Study Environment', 'Coping_Mechanism',
       'Lifestyle_Habits', 'Academic Competition', 'Academic_Stress_Level',
       'year', 'day_name', 'day', 'Month_Name'],
      dtype='object')
```

[67]:
```
# Let's assume we want to study the relationship between the study environment␣
↪and Lifestyle_Habits.
contingency_table_three = pd.crosstab(df1['Study Environment'],␣
↪df1['Lifestyle_Habits'],normalize='index')*100
```

```
print(contingency_table_three)
```

```
Lifestyle_Habits    Decline         No        Yes
Study Environment
Noisy              3.125000  87.500000   9.375000
Peaceful           5.797101  91.304348   2.898551
disrupted          5.263158  81.578947  13.157895
```

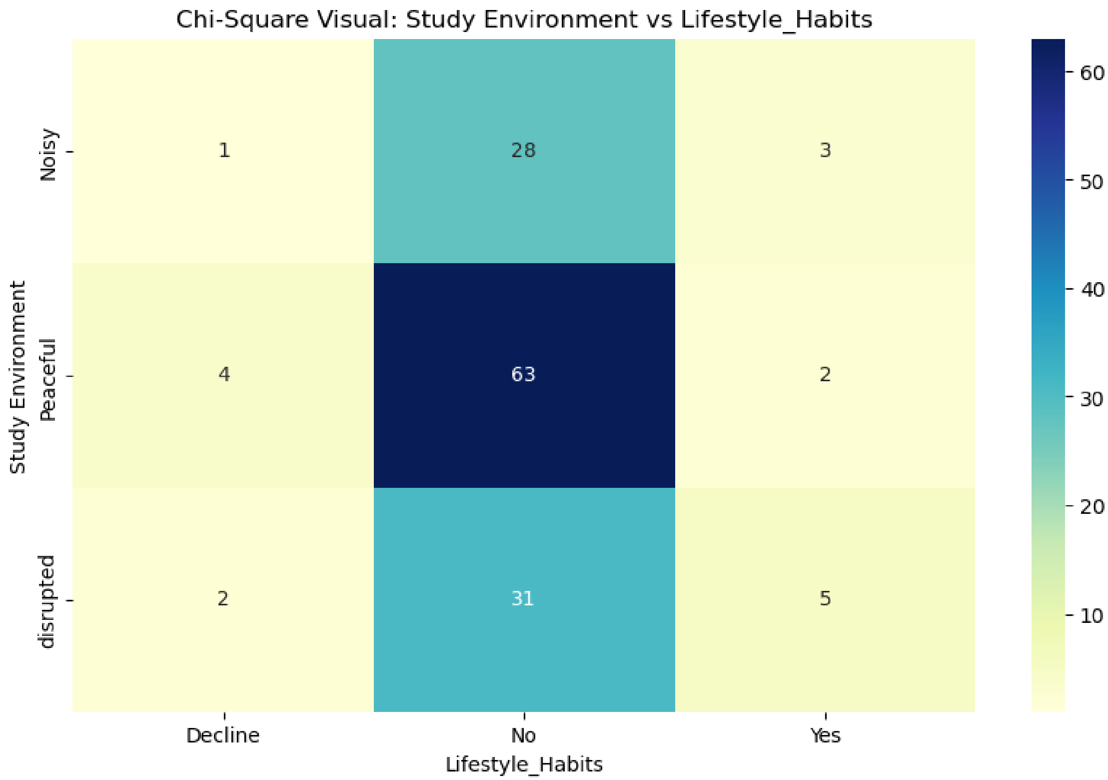
[68]:

```
# Chi-Square
chi3, p3, dof3, expected3 = chi2_contingency(contingency_table_three)

print(f"Chi-Square Statistic: {chi3}")
print(f"P-value: {p3}")
```

```
Chi-Square Statistic: 7.749959724904954
P-value: 0.10117901666956257
```

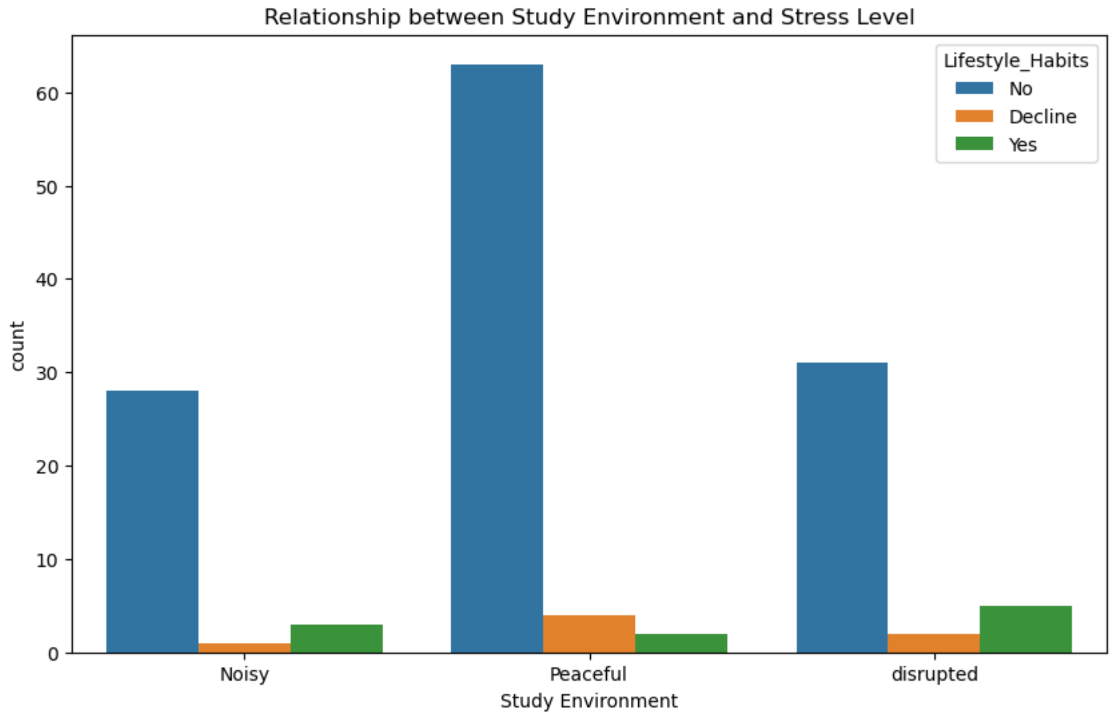
[69]:

```
# 1. (Contingency Table)
ct3 = pd.crosstab(df1['Study Environment'], df1['Lifestyle_Habits'])

# 2.              (Heatmap)
plt.figure(figsize=(10, 6))
sns.heatmap(ct3, annot=True, cmap='YlGnBu', fmt='d')

plt.title('Chi-Square Visual: Study Environment vs Lifestyle_Habits')
plt.xlabel('Lifestyle_Habits')
plt.ylabel('Study Environment')
plt.show()
```



[70]:

```
plt.figure(figsize=(10,6))
sns.countplot(data=df1, x='Study Environment', hue='Lifestyle_Habits')
plt.title('Relationship between Study Environment and Lifestyle_Habits')
plt.show()
```



6.0.7 How do coping mechanisms differ between high school students and university students?"

[73]:
```
df1.columns
```

[73]:
```
Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
       'Academic pressure home', 'Study Environment', 'Coping_Mechanism',
       'Lifestyle_Habits', 'Academic Competition', 'Academic_Stress_Level',
       'year', 'day_name', 'day', 'Month_Name'],
      dtype='object')
```

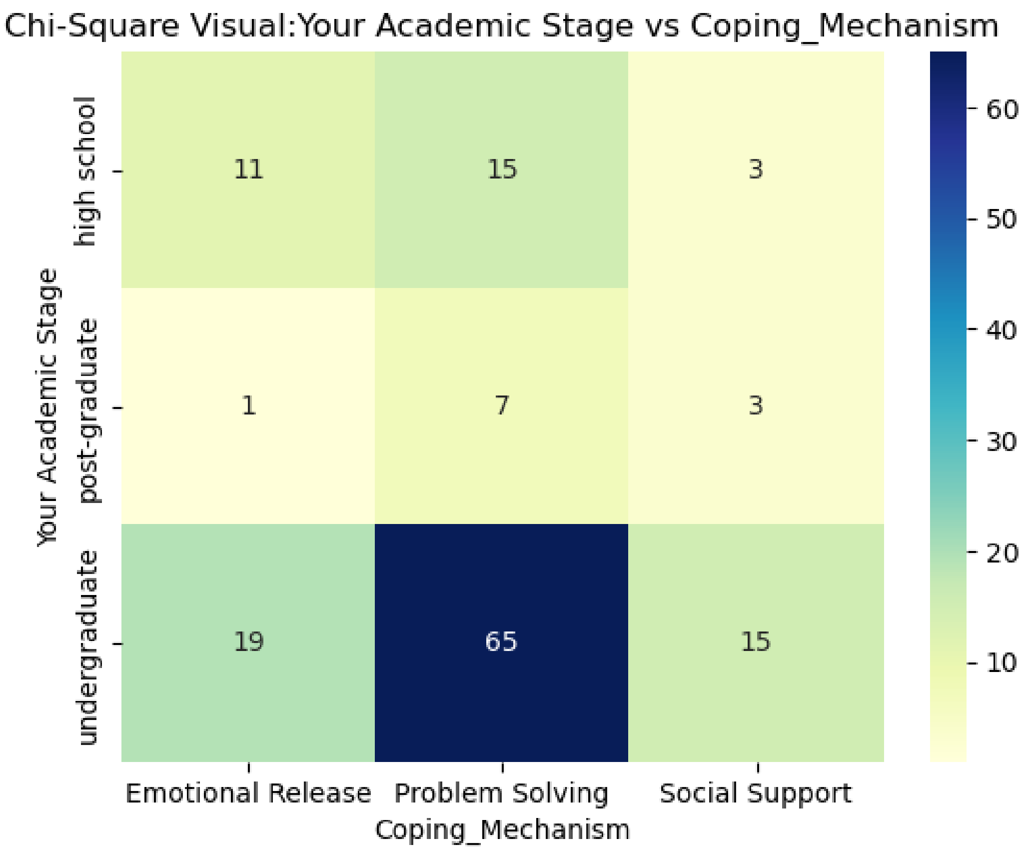
[81]:
```
contingency_table_4 = pd.crosstab(df1["Your Academic␣
↪Stage"],df1["Coping_Mechanism"],normalize ="index")*100
print(contingency_table_4)
```

```
Coping_Mechanism     Emotional Release  Problem Solving  Social Support
Your Academic Stage
high school                  37.931034        51.724138       10.344828
post-graduate                 9.090909        63.636364       27.272727
undergraduate                19.191919        65.656566       15.151515
```

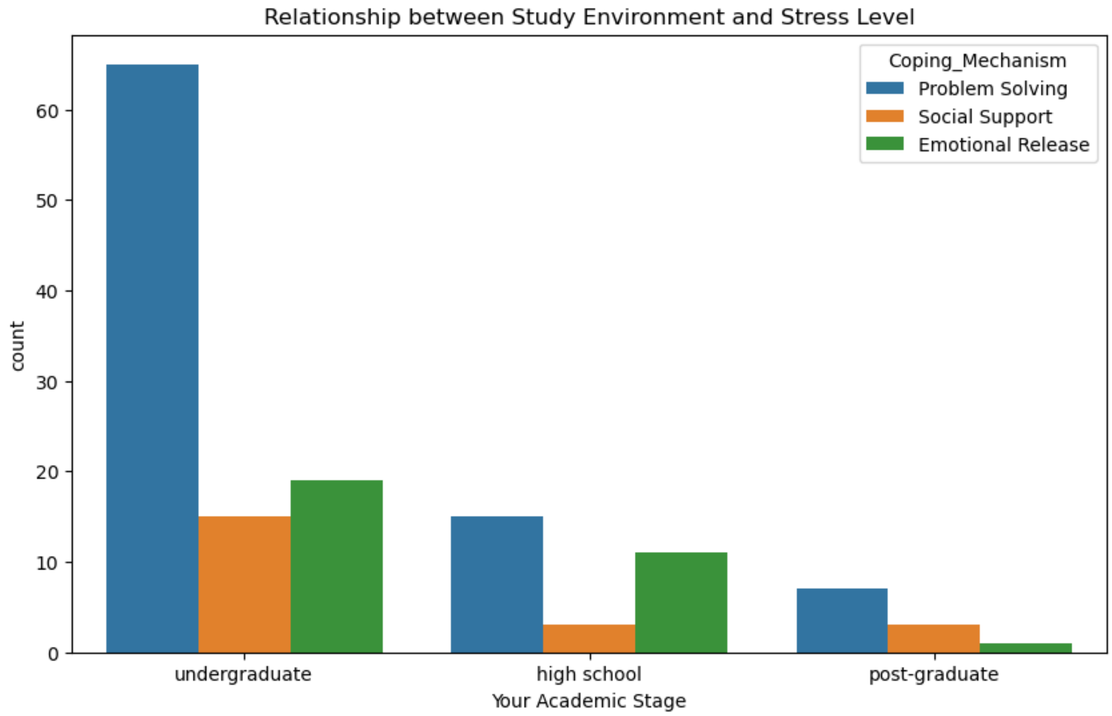
[100]:
```
chi4 , p4 , dof4 , expected4 = chi2_contingency(contingency_table_4)
print(f"Chi-Square Statistic: {chi4}")
print(f"P-value: {p4}")
```

```
Chi-Square Statistic: 273.1294809703369
P-value: 9.084476463079819e-49
```

[102]:
```
ct4 =  pd.crosstab(df1["Your Academic Stage"],df1["Coping_Mechanism"])

sns.heatmap(ct4,annot=True,cmap='YlGnBu',fmt='d')
plt.title('Chi-Square Visual:Your Academic Stage vs Coping_Mechanism')
plt.xlabel('Coping_Mechanism')
plt.ylabel('Your Academic Stage')

plt.show()
```



[103]:
```
plt.figure(figsize=(10,6))
sns.countplot(data=df1, x='Your Academic Stage', hue='Coping_Mechanism')
plt.title('Relationship between Your Academic Stage and Coping_Mechanism')
plt.show()
```



[ ]:

6.0.8 Do students facing high peer pressure experience overwhelming stress levels more than others?"

[104]:
```
df1.columns
```

[104]:
```
Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
       'Academic pressure home', 'Study Environment', 'Coping_Mechanism',
       'Lifestyle_Habits', 'Academic Competition', 'Academic_Stress_Level',
       'year', 'day_name', 'day', 'Month_Name'],
      dtype='object')
```

[105]:
```
contingency_table_5 = pd.crosstab(df1["Peer␣
 ↪pressure"],df1["Academic_Stress_Level"],normalize ="index")*100
print(contingency_table_5)
```

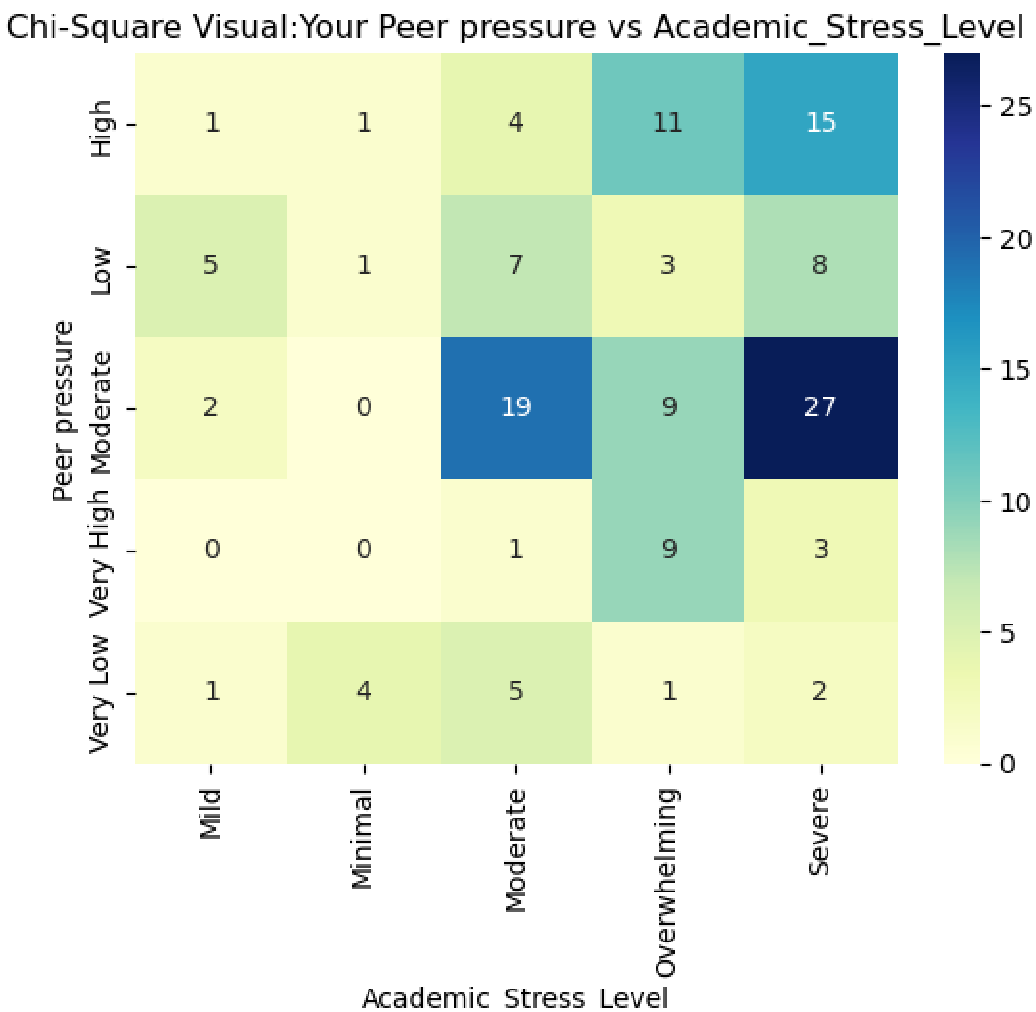
```
Academic_Stress_Level       Mild    Minimal   Moderate  Overwhelming  \
Peer pressure
High                    3.125000   3.125000  12.500000     34.375000
Low                    20.833333   4.166667  29.166667     12.500000
Moderate                3.508772   0.000000  33.333333     15.789474
Very High               0.000000   0.000000   7.692308     69.230769
Very Low                7.692308  30.769231  38.461538      7.692308
```

```
Academic_Stress_Level     Severe
Peer pressure
High                   46.875000
Low                    33.333333
Moderate               47.368421
Very High              23.076923
Very Low               15.384615
```

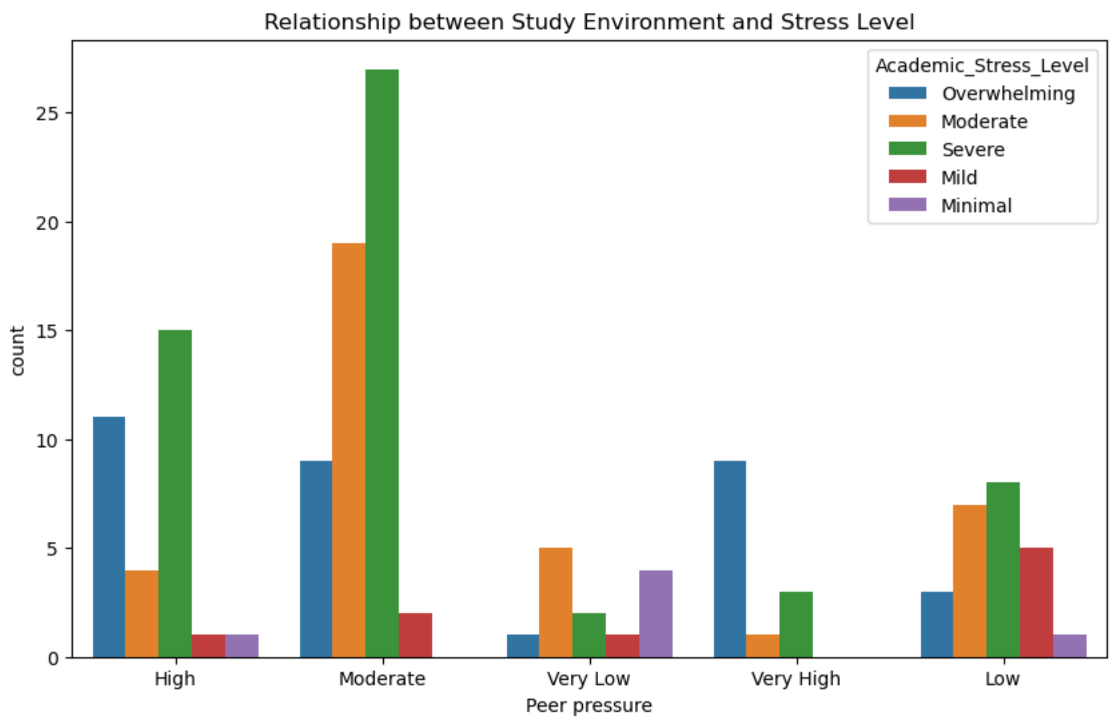
[106]:
```
chi5 , p5 , dof5 , expected5 = chi2_contingency(contingency_table_5)
print(f"Chi-Square Statistic: {chi5}")
print(f"P-value: {p5}")
```

```
Chi-Square Statistic: 273.1294809703369
P-value: 9.084476463079819e-49
```

[110]:
```
ct5 =  pd.crosstab(df1["Peer pressure"],df1["Academic_Stress_Level"])

sns.heatmap(ct5,annot=True,cmap='YlGnBu',fmt='d')
plt.title('Chi-Square Visual:  Peer pressure vs Academic_Stress_Level')
plt.xlabel('Academic_Stress_Level')
plt.ylabel('Peer pressure')

plt.show()
```



[111]:
```
plt.figure(figsize=(10,6))
sns.countplot(data=df1, x='Peer pressure', hue='Academic_Stress_Level')
plt.title('Relationship between Peer pressure and Academic_Stress_Level')
plt.show()
```




```
[ ]:
```

6.0.9 "Which academic level records the highest levels of academic competition?"

```
[112]: df1.columns
```

```
[112]: Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
              'Academic pressure home', 'Study Environment', 'Coping_Mechanism',
              'Lifestyle_Habits', 'Academic Competition', 'Academic_Stress_Level',
              'year', 'day_name', 'day', 'Month_Name'],
             dtype='object')
```

```
[ ]:
```

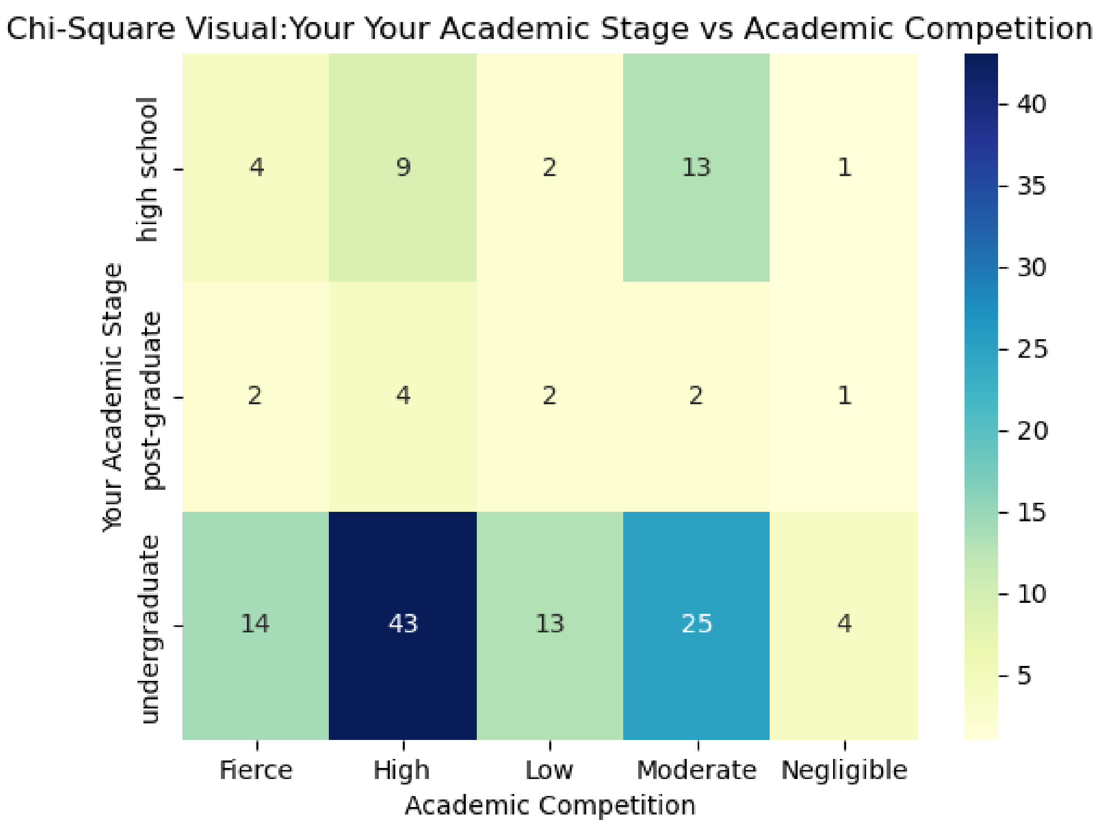
```
[115]: contingency_table_6 = pd.crosstab(df1["Your Academic Stage"],df1["Academic␣
         ↪Competition"],normalize ="index")*100
       print(contingency_table_6)
```

```
Academic Competition     Fierce       High        Low   Moderate  Negligible
Your Academic Stage
high school          13.793103  31.034483   6.896552  44.827586    3.448276
post-graduate        18.181818  36.363636  18.181818  18.181818    9.090909
undergraduate        14.141414  43.434343  13.131313  25.252525    4.040404
```

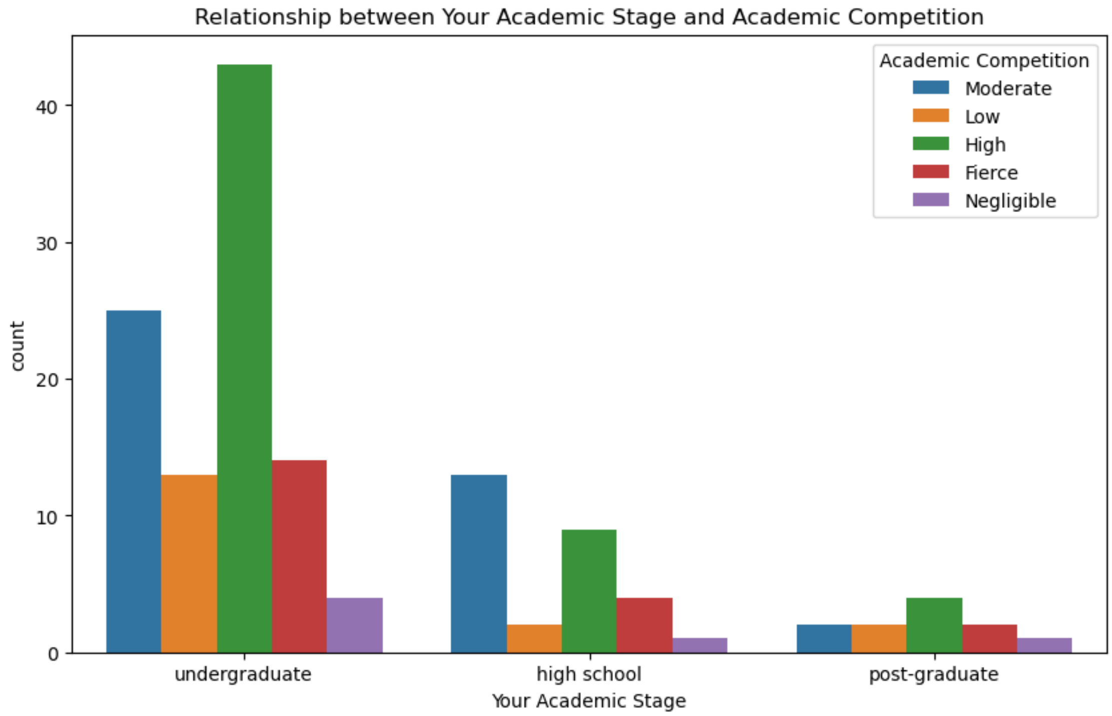
[116]:
```
chi6 , p6 , dof6 , expected6 = chi2_contingency(contingency_table_6)
print(f"Chi-Square Statistic: {chi6}")
print(f"P-value: {p6}")
```

```
Chi-Square Statistic: 24.319024961898315
P-value: 0.002025781747152824
```

[119]:
```
ct6 =  pd.crosstab(df1["Your Academic Stage"],df1["Academic Competition"])

sns.heatmap(ct6,annot=True,cmap='YlGnBu',fmt='d')
plt.title('Chi-Square Visual: Your Academic Stage vs Academic Competition')
plt.xlabel('Academic Competition')
plt.ylabel('Your Academic Stage')

plt.show()
```



[118]:
```
plt.figure(figsize=(10,6))
sns.countplot(data=df1, x='Your Academic Stage', hue='Academic Competition')
plt.title('Relationship between Your Academic Stage and Academic Competition')
plt.show()
```



[121]:
```
df1.columns
```

[121]:
```
Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
       'Academic pressure home', 'Study Environment', 'Coping_Mechanism',
       'Lifestyle_Habits', 'Academic Competition', 'Academic_Stress_Level',
       'year', 'day_name', 'day', 'Month_Name'],
      dtype='object')
```

6.0.10 "Are there specific days of the week when students tend to report higher stress levels?"

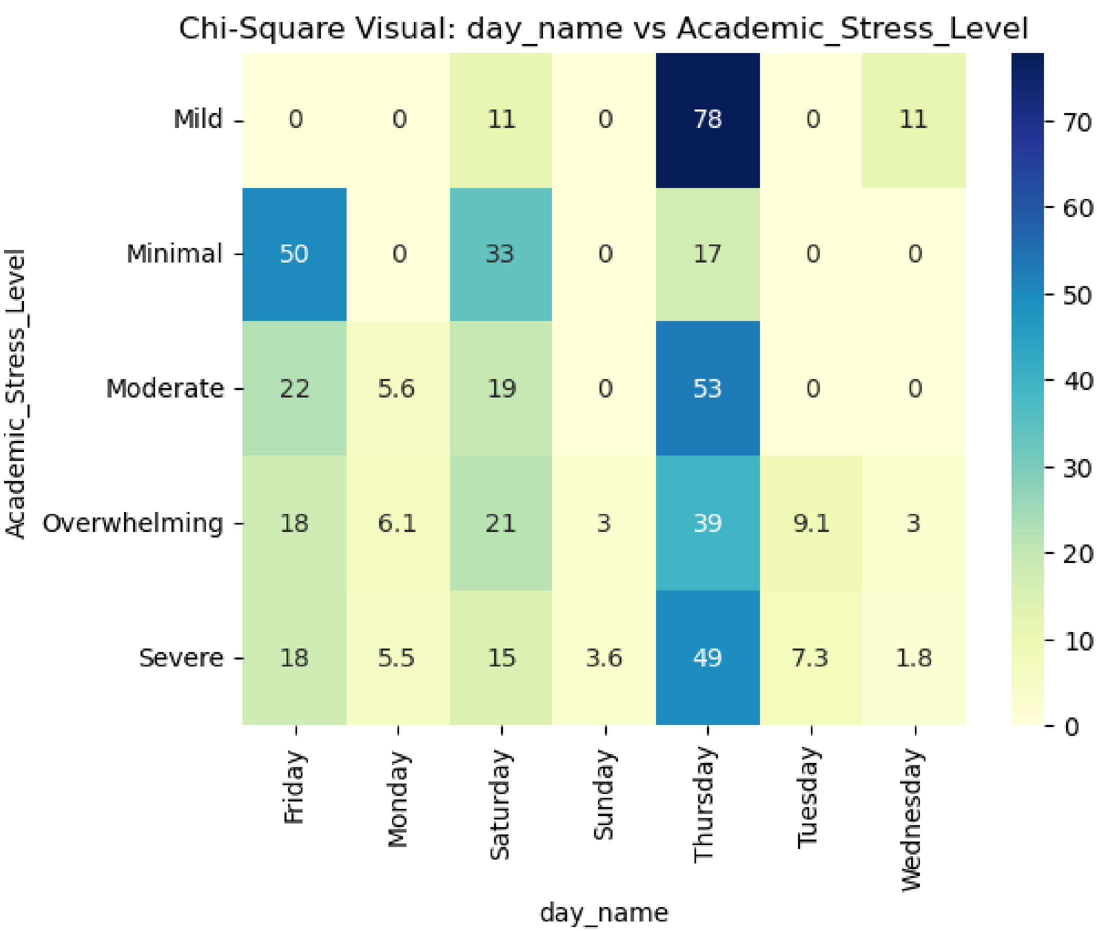
[124]:
```
contingency_table_7 = pd.
 ↪crosstab(df1["Academic_Stress_Level"],df1["day_name"],normalize ="index")*100
print(contingency_table_7)
```

```
day_name                  Friday    Monday   Saturday    Sunday   Thursday  \
Academic_Stress_Level
Mild                    0.000000  0.000000  11.111111  0.000000  77.777778
Minimal                50.000000  0.000000  33.333333  0.000000  16.666667
Moderate               22.222222  5.555556  19.444444  0.000000  52.777778
Overwhelming           18.181818  6.060606  21.212121  3.030303  39.393939
Severe                 18.181818  5.454545  14.545455  3.636364  49.090909

day_name                 Tuesday  Wednesday
```

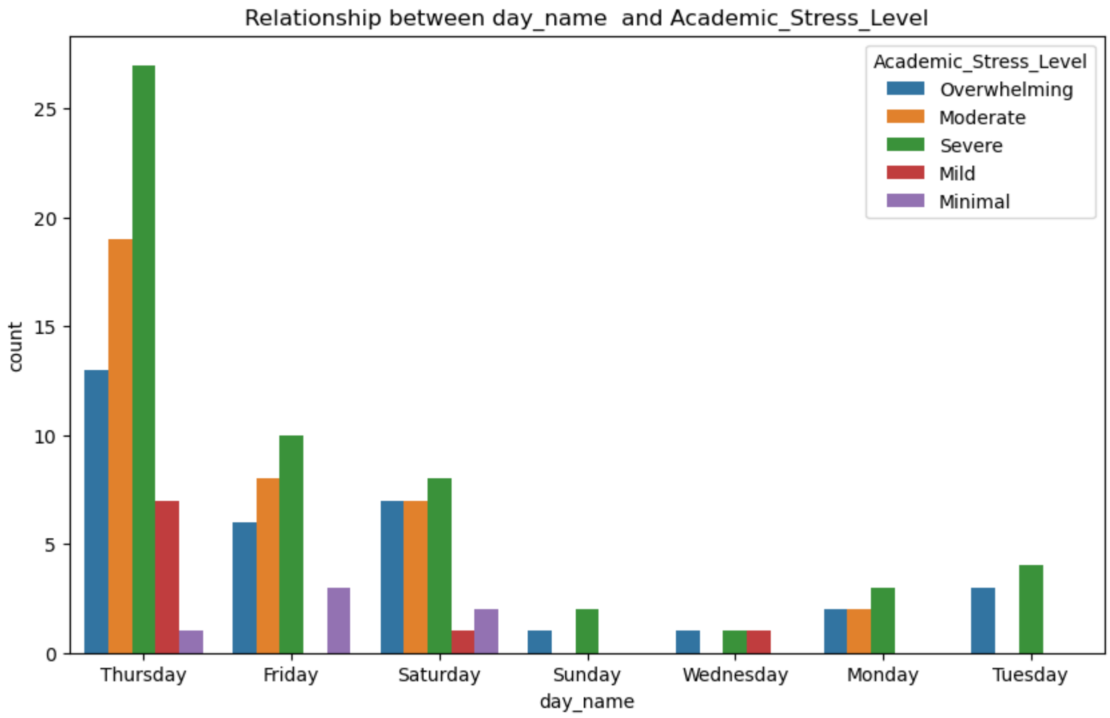
```
Academic_Stress_Level
Mild                    0.000000  11.111111
Minimal                 0.000000   0.000000
Moderate                0.000000   0.000000
Overwhelming            9.090909   3.030303
Severe                  7.272727   1.818182
```

[ ]:
```
chi7 , p7 , dof7 , expected7 = chi2_contingency(contingency_table_7)
print(f"Chi-Square Statistic: {chi7}")
print(f"P-value: {p7}")
```

[126]:
```
ct7 = pd.crosstab(df1["Academic_Stress_Level"],df1["day_name"],normalize␣
↪="index")*100

sns.heatmap(ct7,annot=True,cmap='YlGnBu')
plt.title('Chi-Square Visual: day_name vs Academic_Stress_Level')
plt.xlabel('day_name')
plt.ylabel('Academic_Stress_Level')

plt.show()
```



[127]:
```
plt.figure(figsize=(10,6))
sns.countplot(data=df1, x='day_name', hue='Academic_Stress_Level')
plt.title('Relationship between day_name  and Academic_Stress_Level')
plt.show()
```

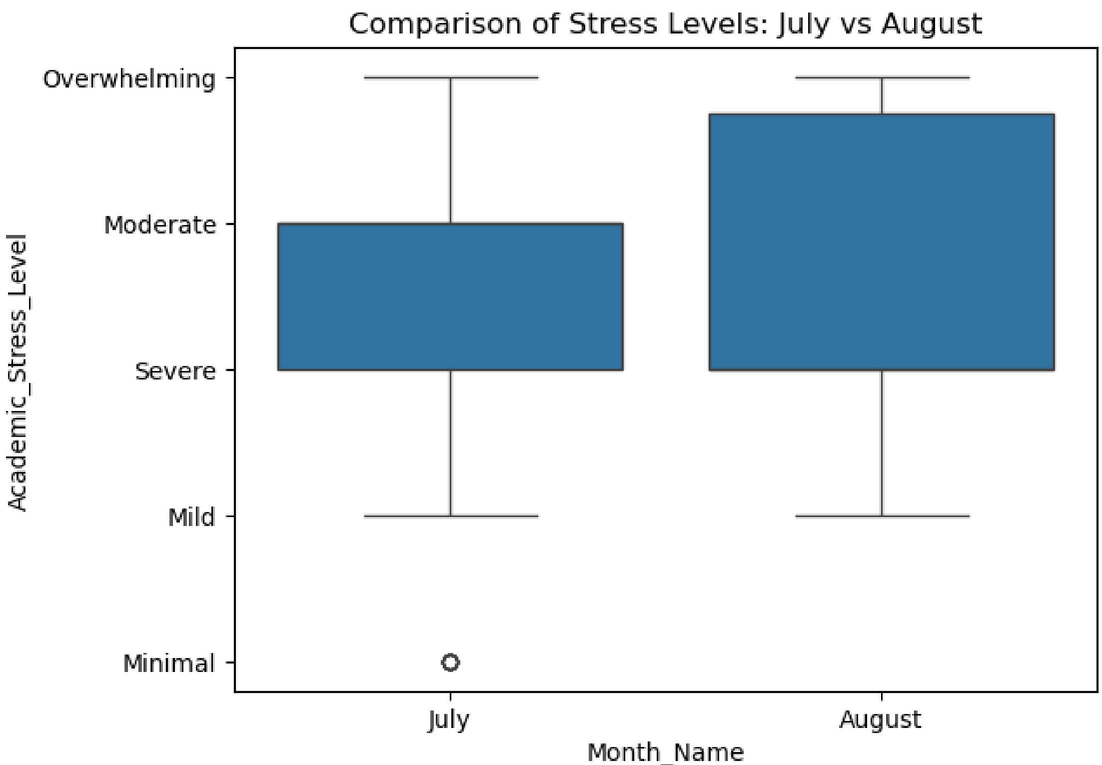


[129]:
```
df1.columns
```

[129]:
```
Index(['Timestamp', 'Your Academic Stage', 'Peer pressure',
       'Academic pressure home', 'Study Environment', 'Coping_Mechanism',
       'Lifestyle_Habits', 'Academic Competition', 'Academic_Stress_Level',
       'year', 'day_name', 'day', 'Month_Name'],
      dtype='object')
```

6.0.11 ”Did the average stress level distributed among students differ between July and August?

[135]:
```
sns.boxplot(x='Month_Name', y='Academic_Stress_Level',␣
 ↪data=df1[df1['Month_Name'].isin(['July', 'August'])])
plt.title('Comparison of Stress Levels: July vs August')
plt.show()
```



[143]:
```
stress_map = {
    'Minimal': 1,
    'Mild': 2,
    'Moderate': 3,
    'Severe': 4,
    'Overwhelming': 5
}

df1['stress_score'] = df1['Academic_Stress_Level'].map(stress_map)
```

[150]:
```
df1['stress_score']= df1['stress_score'].astype("int")
```

[151]:
```
df1.sample()
```

[151]:
```
                   Timestamp Your Academic Stage Peer pressure  \
73 2025-07-25 10:43:24+00:00       undergraduate      Moderate

   Academic pressure home Study Environment Coping_Mechanism Lifestyle_Habits  \
73      Moderate Pressure         disrupted  Problem Solving               No

   Academic Competition Academic_Stress_Level  year day_name day Month_Name  \
73                 High                Severe  2025   Friday  25       July
```

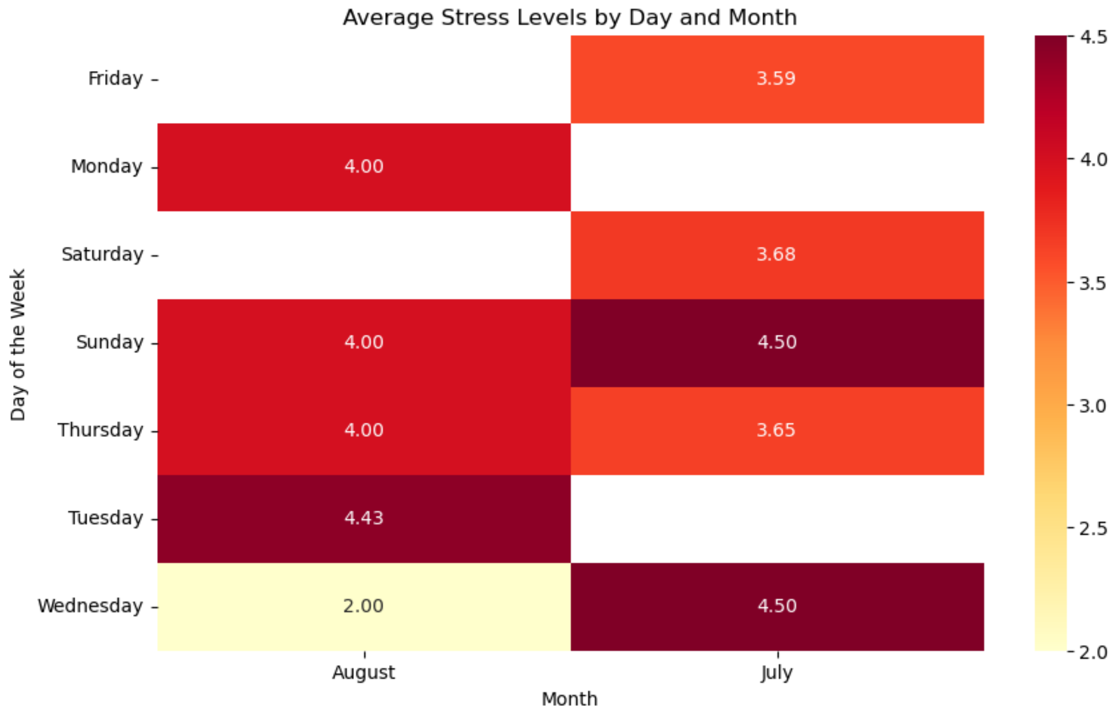
```
    stress_score
73             4
```

[154]:
```
july_stress = df1[df1['Month_Name'] == 'July']['stress_score']
august_stress = df1[df1['Month_Name'] == 'August']['stress_score']
```

[155]:
```
from scipy import stats
#  T-test
t_stat, p_value = stats.ttest_ind(july_stress, august_stress, nan_policy='omit')


print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.4f}")
```

```
T-statistic: -1.4820
P-value: 0.1406
```

[158]:
```
pivot_table = df1.pivot_table(index='day_name',
                             columns='Month_Name',
                             values='stress_score',
                             aggfunc='mean')

plt.figure(figsize=(10, 6))
sns.heatmap(pivot_table, annot=True, cmap='YlOrRd', fmt=".2f")

plt.title('Average Stress Levels by Day and Month')
plt.xlabel('Month')
plt.ylabel('Day of the Week')
plt.show()
```


Average Stress Levels by Day and Month
Friday
Monday
Saturday
Sunday
Thursday
Tuesday
Wednesday
Day of the Week
3.59
4.00
3.68
4.00
4.50
4.00
3.65
4.43
2.00
4.50
August
July
Month
4.5
4.0
3.5
3.0
2.5
2.0

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